MIL-C-42005A (DM)
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MILITARY SPECIFICATION

CHAIR, DENTAL, PATIENT

This specification is approved for use by the Defense Personnel Support Center based upon currently available technical information but it has not been approved for promulgation as a revision of MIL-C-42005. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers a lounge type, self-powered, adjustable dental operating chair suitable for use in the performance of the treatment of dental patients in the "sit-down" and "stand-up" dentistry.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document; should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

CCC-A-680	-Artificial Leather, (Cloth Coated), Vinyl Resin, Expanded Layer, (Upholstery).

MILITARY

MIL-M-7298	-Manual, Technical, Commercial Equipment.

STANDARDS

FEDERAL

FED-STD-H28	-Screw-Thread Standards for Federal Services.
FED-STD-191A	-Textile Test Methods, Method 5903.

MILITARY

MIL-STD-1623 (SH)	-Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Finish (Naval Shipboard Use).
MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-889	-Dissimilar Metals

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the non-Government documents which are current on the date of the solicitation.

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AMERICAN NATIONAL STANDARDS INSTITUTE/AMERICAN DENTAL ASSOCIATION(ANSI/ADA)

ANSI/ADA No. 46 -Specification for Dental Chairs

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

UNDERWRITERS' LABORATORY, INC. (UL)

UL 544 - Standard for Safety Medical and Dental Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I., NY 11746, 207 East Ohio Street, Chicago, IL 60611, 333 Pfingsten Road, Northbrook, IL 60062 and 1655 Scott Boulevard, Santa Clara, CA 95050).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein. When materials are not specifically designated, those used shall be of the same or better quality as used in commercial practice, and entirely suitable for the purpose intended. No wood or wood products shall be used in construction except as a portion of the replaceable upholstery.

3.1.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. Where such contact cannot be avoided, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

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3.2 General requirements. The chair shall comply with the applicable requirements of American National Standards Institute/American Dental Association (ANSI/ADA) Specification No. 46 for class A Chairs, Underwriters' Laboratories Inc. (UL) 544 Standard for Safety Medical and Dental Equipment, and additional requirements and exceptions as specified herein.

3.2.1 First article. When specified in the contract or purchase order, a sample shall be subjected to the first article inspection (see 4.3 and 6.1.1).

3.2.2 Standard commercial product. The chair shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the chair being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, or manufacturer's catalogs or brochures, and represents the latest production model.

3.2.3 Additional safety requirements.

3.2.3.1 The chair shall be equipped with safety limit switches at all extremes of travel under powered operation.

3.2.3.2 The chair shall provide unrestricted access to electrical connections and drive mechanism for the purpose of maintenance and repair. All openings to access areas shall be provided with a suitable cover to prevent shock or safety hazards to the operator and the patient during normal operations.

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3.2.4 Fire and casualty hazards. The contractor shall provide proof, to the procurement contracting officer, not less than thirty days prior to delivery, that the chairs to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories, Inc. (UL) Standard of Safety 544. A copy of the test report shall be made available to the Government inspection personnel for their use in determining that the chairs being submitted for acceptance are identical to the chair submitted for approval. Proof shall consist of one of the following:

a. UL listing and UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, and the Government inspection personnel, upon request.

b. A label or equivalent means of identification affixed to the product as evidence that the product design has been examined and tested for design safety compliance by a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL Standards.

The Government reserves the right to require a facilities report and to conduct a facilities survey. In support of the label/product identification, the independent testing laboratory shall provide surveillance over the product manufacturing system to the degree necessary to insure that production units are manufactured in the same manner and with the same type of components as the test sample evaluated and upon which the validity of the product identification is based. The testing laboratory shall develop a detailed test report which includes a conclusion as to whether the test sample evaluated is "approved" or "not approved" as complying with all requirements of the standards of safety.

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3.3 Design, construction, and dimensions. The chair shall be a lounge type, self-powered, adjustable dental operating chair suitable for positioning a patient for both "sit-down" or "stand-up" in the performance of dental procedures in a safe and efficient manner. The chair shall consist of the following major assemblies or components: floor plate, lift base mechanism, controls, and a chair top assembly including a seat, backrest, headrest, armrests, and toe board. The chair shall have a rigid metal frame supporting the backrest, seat, and toe board. The chair shall be properly contoured and cushioned for patient comfort. The chair shall provide maximum accessibility to the patient's oral cavity. The chair shall be compatible and provisioned to accept an adaptor for the system operating unit (chair side mounted over the patient); and dental operatory light, when specified (see 6.1.1).

3.3.1 Floor plate. The floor plate shall provide a solid foundation for the entire chair including all accessories under all load conditions. The floor plate shall be sufficient in area so as to prevent rocking or tipping and shall enhance stability, but shall not interfere with stool castors. The floor plate shall be protected on both sides (top and bottom) with a corrosion and chip-resistant finish.

3.3.2 Chair mechanism. All hydraulically operated movements shall be capable of speed adjustments. All pivots, bushings, and bearings shall be self lubricating.

3.3.3 Chair top assembly.

3.3.3.1 The chair shall facilitate "sit-down" dentistry. The width of the backrest shall be tapered from the seat to the headrest to allow unhampered access to the operating area during treatment. The thickness of the upper half of the backrest shall not exceed 1-1/2 inches with the backrest cushion removed nor shall it exceed 4 inches with the backrest cushion in place. The width of the backrest shall not exceed 15 inches at 6 inches below the top of the backrest.

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3.3.3.2 The seat shall have a minimum width of 18 inches between armrests. The chair shall have a minimum overall length of 70 inches, when measured from the top of the headrest to the end of the toeboard with the chair in the supine position and the headrest fully retracted (as specified in 4.4.5.2.1).

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3.3.3.3 The armrests shall fully support the patient's arms, shall be movable to permit patient access from either side of the chair and shall feature a positive locking mechanism. The armrests shall be durably and sturdily constructed such that each shall withstand a vertical load of 100 pounds when tested as specified in 4.4.5.2.10 without failure or permanent impairment of function as specified in ANSI/ADA Specification No.46. Each armrest shall facilitate the performance of phlebotomy procedures.

3.3.3.4 The chair shall be available with a variety of headrests suitable for general dentistry, prosthodontic and oral surgical procedures. Each type headrest shall be adjustable in height, and positive locking in each selected position. The headrest shall be durably and sturdily constructed and withstand a 70 pound vertical force when tested as specified in 4.4.5.2.9. The headrest shall be the following types, as specified (see 6.1.2).

a. Adjustable height, positive locking, articulating headrest.
b. Adjustable height, positive locking, non-articulating headrest.
c. Adjustable height, positive locking, prosthodontic (prosthetic) headrest.

3.3.4 Chair positions, movements and adjustments.

3.3.4.1 The seat height shall be fully adjustable by a self-powered mechanism such that the lowest position shall be no greater than 18 inches and the highest position shall be no less than 28 inches so as to accommodate both "sit-down" and "stand-up" dentistry, when measured as specified in 4.4.5.2.4. The chair shall be stable, rigid, and immovable in all operating positions so as to facilitate the performance of delicate dental procedures.

3.3.4.2 The chair shall be equipped with a base that shall permit easy rotation of the chair top assembly of 30 degrees (minimum) in either direction (60 degrees total, minimum) in the horizontal plane with a positive locking mechanism at any point within the full range of motion when measured as specified in 4.4.5.2.5. The base shall resist denting, marring and chipping.

3.3.4.3 The chair shall be capable of positioning the patient in the modified Trendelenberg position of no less than 5 degrees below the horizontal when measured as specified in 4.4.5.2.7.

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3.3.4.4 The chair shall be capable of positioning the patient in an upright position of no less than 60 degrees above the horizontal when measured as specified in 4.4.5.2.3.

3.3.4.5 The backrest shall be self-powered and shall have manual controls to allow positioning of the backrest and seat at any intermediate position throughout the range of motion. Shall be capable of at least one selectively programmable operating position and a patient return position with emergency stop. The transition from automatic to manual mode and vice versa shall be immediate and shall require no operational steps. All controls shall be easy to operate and shall be equally accessible from either side of the chair. A chair lift mechanism design which poses the potential for objects to be crushed beneath it shall be equipped with a safety shut-off mechanism.

3.3.4.6 The chair shall be capable of operation from the fully reclined to the fully upright position or vice versa, and from the fully lowered to the fully raised position or vice versa in 15 seconds or less when tested as specified in 4.4.5.2.6.

3.3.5 Toe board or seat/toe board assembly.

3.3.5.1 The toe board or seat/toe board assembly shall be self-powered and move in coordination and synchronization with the movement of the backrest so as to optimize patient comfort. All operations shall be quiet and smooth so as to be free of disconcerting noise and sudden or jerky movement.

3.3.5.2 The toe board shall be capable of supporting a 220 pound load at the end of the board without structural damage or tipping of the chair, when tested as specified in 4.4.5.2.2.

3.3.6 Controls.

3.3.6.1 The chair shall be equipped with a foot control to permit aseptic operation of the dental chair. The foot control shall duplicate all functions of the manual controls and shall be easily movable so as to permit equivalent left or right handed operation. All controls shall operate at a maximum of 24 volts.

3.3.7 Upholstery.

3.3.7.1 The upholstery shall be smooth vinyl with flush seams. Upholstery shall be field replaceable by each separate section. All upholstery which is permanently fastened to a solid support backing shall be edgewrapped.

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The vinyl covering material, including cloth backing, shall be no less than 0.04 inch thick. Upholstery material shall meet the flammability requirements of CCC-A-680a (Federal Standard 191A, Test Method 5903 shall be applicable). The seat cushion shall be reversible when specified, (see 6.1.2).

In addition, for Navy requisitions, the upholstery including cushion padding shall meet the fire performance requirements of MIL-STD-1623 (SH) for Naval Shipboard use, (see 6.1.1 and procurement documents). The upholstery shall be neatly cut, sewed, and sealed where applicable. The upholstery shall neatly and uniformly surround the particular component with sufficient tension to maintain a smooth appearance before and after use. The upholstery shall not contain any cuts, tears, rips, punctures, or any defects which shall impair serviceability or detract from appearance.

3.3.7.2 The upholstery of the toe board shall be protected with a removable scuff cover.

3.3.7.3 The upholstery shall be of the following colors, as specified, (see 6.1.2).

a. tan/beige	e. light blue
b. brown	f. gray
c. rust	g. green
d. dark blue	h. other

3.3.8 Electrical characteristics. The electrical power requirements shall be as follows, as specified (see 6.1.2.)

a. 115 $\pm$ 10% volts, 60 $\pm$ 2 hertz.
b. 115 $\pm$ 10% volts, 50 $\pm$ 2 hertz.
c. 230 $\pm$ 10% volts, 50 $\pm$ 2 hertz.
d. 230 $\pm$ 10% volts, 60 $\pm$ 2 hertz.

The chair shall be furnished with a power cable of sufficient length to accommodate a normal operating system terminating with a three prong hospital grade plug.

3.3.9 Overall operation of chair. The assembled chair (all operating components, adjustments, and controls) shall operate quietly, smoothly, efficiently, shall show no signs of defects or malfunctioning, and shall be capable of moving through positions described herein, when tested as specified in 4.4.5.2.8.

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3.4 Finish. All surfaces shall be clean and free of foreign matter, flux, or welding residue. The finish of components shall be as specified. If no finish is specified, the finish shall be entirely suitable for the purpose intended. Finish on edges and surfaces shall be uniform and free from burrs, slivers, sharp edges, cracks, or other defects. All exposed surfaces of the light shall be smooth and made of materials which can withstand repeated wipe down with ADA acceptable disinfectants without deterioration of function or esthetics.

3.4.1 Plating (coatings). Platings and coatings shall be uniform in thickness and appearance; shall be free from blisters, peeling, visible porosity, or other defects. The base metal to be plated shall be free from flaws or defects which will be detrimental to the final finish and shall be subjected to such undercoating, polishing, cleaning, dipping and plating procedures as are necessary to yield deposits with the desired function, quality and appearance.

3.5 Workmanship. Workmanship shall be in accordance with best commercial practice. All components shall be clean, well finished, and free from defects which detract from their appearance or impair their serviceability.

3.5.1 Stitching and seams. Seams shall be properly sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams shall result.

3.5.2 Heat seals. Heat seals (where applicable) shall be neat, uniform and provide a sound and strong seam. The heat seal seam shall not be blistered, cracked, creased, or wrinkled. The seam shall be continuous and provide a complete secure seam throughout the entire seam area.

3.5.3 Castings. Castings shall be sound and free from repairs, warping, or other defects that may render the casting unsound for use. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste shall not be permitted.

3.5.4 Welding, brazing and soldering. All welds, brazing, and soldering shall be durable and sound in every detail. Welding shall be in accordance with the latest practices recommended by the American Welding Society. Components shall be joined without any resultant distortion and shall be free from twists or strains.

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All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All aluminum parts shall be welded within 48 hours after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour time period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall he added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks.

Exposed welds shall be smooth and blend with adjacent surfaces. All welds shall be cleaned, discoloration removed, and be free from slag and spatter before finishing. Welds shall be ground, only to the degree necessary to remove sharp edges or projections which may cause defects to adjacent surfaces or can be injurious to the touch.

3.5.5 Machine work. Machining shall be in accordance with best commercial practice. All bolt, machine screw and rivet holes shall be accurate and clean. The holes for screws and bolts shall be sized appropriately for the size screw or bolt used. All bolts, nuts, screws, and fasteners shall be completely fastened and tight.

3.5.6 Metal fabrication. Metals (sheets, plates and tubing) shall be free from dents, folds, or distortions. Material shall be straightened by methods that will not cause injury to the metal. Operations (such as bending, shearing, or cutting) shall be done neatly and accurately. Metal forming shall be made with controlled means to ensure uniformity of size and shape.

3.5.7 Hardware. All hardware shall be entirely suitable for the purpose intended and shall be completely compatible with the material to which it is attached or which it joins, including plated material. Any fastening device required to be loosened, tightened, or removed for repair, maintenance or adjustment shall not be swaged, peened, staked or otherwise deformed to prevent loosening of the device. All rotative fasteners shall be correctly torqued. All threaded parts shall comply with Federal Standard H28.

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3.6 Approval of colors. Samples of the upholstery for color and texture shall be submitted to the contracting officer for approval prior to commencing production, (see 6.1.1).

3.7 Hydraulic charge. Chairs utilizing hydraulic fluid shall have the hydraulic system charged so that the chair shall be in the ready-to-use condition.

3.8 Identification marking. The chair shall have one or more nameplates, permanently fastened and reasonably accessible, containing the name of manufacturer, manufacturer's type and model, serial number and electrical characteristics. The serial number shall be traceable by the manufacturer to the date of manufacture.

3.9 Interchangeability. All parts having the same manufacturer's part number shall be functionally, and dimensionally interchangeable.

3.10 Tolerances. Where tolerances are not specified, standard commercial tolerances shall be permissible. Tolerances shall be held to a minimum where accuracy is of paramount importance for proper operation and interchangeability of component parts. Assemblies shall be precision-fitted for durable, smooth, and consistent functioning.

3.11 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual (installation manual and maintenance/repair manual, and an operations manual) which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298.

The installation manual and maintenance/repair manual shall include the following: illustrated step by step installation instructions; list of special tools (if required); exploded drawings with parts identified by part number, functional test procedures; electrical and utility requirements; electrical and utility schematic diagrams; lubrication schedule, lubrication points and instructions for lubrication; user maintenance schedule; telephone number for technical assistance, and a trouble-shooting diagram.

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The operations manual shall include the following: step by step procedures for the operation of the chair including illustrations showing the location and function of each control; location of control adjustments and procedures for adjustments; instructions for the proper cleaning and disinfection of all exposed surfaces; and a daily functional checklist.

As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center.

The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contrtactor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of materials and components, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the manufacturer or supplier, the Item Identification, the name of the contractor, the contract number, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements. Certificates of quality may be furnished for the following: material composition, hardness, tempers, torque requirements on springs, machining and casting requirements (such as flatness, concentricity, and finish), and plating or coating.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is specified in the contract, the first article inspection shall consist of all examinations specified herein, and all testing specified herein and shall apply to the first article sample only (see 6.1.1). The failure of compliance with any examination or test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

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4.4.1 In-process examination. In-process examination shall be conducted to ascertain that components or assemblies which cannot be readily examined in the end item are in accordance with specification requirements.

4.4.1.1 Visual examination. In-process examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 1.0 for major defects and an AQL of 10 for minor defects. The unit of product shall be the component of an end item or an end item, as applicable.

TABLE I. In-Process classification of defects

	Major	Minor
Welding, Brazing & Soldering		
Missing, incomplete, fractured, not fused, not continuous, not sound.	X	
Burn holes, pits.		X
Rough and not uniform.		X
Not free of scale, rust, flux, grease, foreign matter.		X
Welds have sharp edges or projections which may cause holes, tears, impair the upholstery or cause injury when touched.	X	
Excessive solder on joint (runoff).		X
Hardware		
Threads missing or incomplete,	X	
Threads damaged.		X
Finish		
Finish not as specified.	X	
Surfaces not clean, not free of foreign matter, flux, spatters, or other defects.		X

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TABLE I. In-Process classification of defects (Continued)

	Major	Minor
Damage or defects on exterior or interior surfaces such as burrs, slivers, cracks, sharp edges,which affect function or serviceability.	X	
Damage or defects on exterior or interior surfaces such as burrs, slivers, cracks, sharp edges which do not affect function or serviceability.		X
Finish		
Plating or coating (where applicable) missing (affecting function).	X	
Plating or coating (where applicable) missing (not affecting function).		X
Plating (where applicable) not uniform in appearance.		X
Plating (where applicable) not free from blisters, peeling, visible porosity, or other defects (affecting function or serviceability).	X	
Plating (where applicable) not free blisters, peeling, visible porosity, or other defects (not affecting function or serviceability).		X
Workmanship		
Any component fractured, broken, punctured, torn, bowed, deteriorated, or malformed.	X	
Any component dented or creased.		X
Any component not connected or not joined properly.		X
Fastening device requiring loosening or removal is swaged, peened, staked o otherwise permanently fastened.	X	

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TABLE I. In-Process classification of defects (Continued)

	Major	Minor
Components missing.	X	
Interface fits between components not proper (too loose, too tight, binding).	X	
Components not free from defects which may impair their serviceability.	X	
Components not free from defects which do not impair their serviceability.		X
Castings		
Cast components not sound, not free from porosity, cracks or blowholes.	X	
Cast components repaired, or brazed.	X	
Cast components malformed.	X	
Machine Work		
Holes not accurately and cleanly made.	X	
Coarse machine tool or die marks which affect function or serviceability.	X	
Coarse machine, tool or die marks which do not affect function or serviceability.		X
Components not accurately and smoothly machined on contacting surfaces.	X	
Heat Seals (where applicable)		
Seam not neat and uniform.		X
Seam blistered, cracked, or creased.		X
Seam not continuous or secure.	X	
Seam open.	X	

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TABLE I. In-Process classification of defects (Continued)

	Major	Minor
Metal Fabrication (Sheets, plates and tubing)		
Metal not free from dents, folds, or distortions (seriously affecting appearance or serviceability).	X	
Metal not free from dents, folds or distortions (not seriously affecting appearance or serviceability).		X
Shearing, bending, or cutting not neat and accurate (seriously affecting appearance, serviceability, or can cause injury).	X	
Shearing, bending, or cutting not neat and accurate (not seriously affecting appearance, serviceability, or cannot cause injury).		X
Forming not neatly finished.		X

4.4.2 End item examination.

4.4.2.1 Visual examination. End item examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table II. The inspection level shall be II with an acceptable quality level (AQL) of 1.0 for major defects and an AQL of 10 for minor defects. The unit of product shall be one chair.

Table II. End item classification of defects

	Major	Minor
Design and Construction		
Design and construction of chair not as specified.	X	
Color		
Color of upholstery not as specified.	X	

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Table II. End item classification of defects (continued)

	Major	Minor
Cushioning material		
Not flexible.	X	
Not resilient.	X	
Hardware		
Components such as pins, screws and fasteners missing, broken, stripped, cracked, or deformed.	X	
Fasteners loose.		X
Finish		
Finish not as specified.	X	
Surfaces not clean, not free of foreign matter, flux, spatters, or other defects.		X
Damage or defects on exterior or interior surfaces such as burrs, slivers, cracks, sharp edges, which affect function or serviceability.	X	
Damage or defects on exterior or interior surfaces such as slivers, cracks, sharp edges which do not affect function or serviceability.		X
Finish		
Plating or coating (where applicable) missing (affecting function).	X	
Plating or coating (where applicable) missing (not affecting function).		X
Plating or coating (where applicable) not uniform in appearance.		X

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TABLE II. End item classification of defects (Continued)

	Major	Minor
Finish		
Plating or coating (where applicable) not free from blisters, peeling, visible porosity or other defects (affecting function or serviceability.	X	
Plating or coating (where applicable) not free from blisters, peeling, visible porosity or defects (not affecting function or serviceability).		X
Workmanship		
Any component fractured, broken, punctured, torn, bowed, deteriorated, or malformed.	X	
Any component dented or creased.		X
Any component misplaced or not in proper alignment.	X	
Any component not connected or not joined properly.		X
Fastening device requiring loosening or removal is swaged, peened, staked or otherwise permanently fastened.	X	
Components missing.	X	
Components do not fit or mate properly.	X	
Hydraulic system not charged with hydraulic fluid (if applicable).		X
Workmanship		
Interface fits between components not proper (too loose, too tight/binding).	X	
Components not free from defects which may impair their serviceability.	X	

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TABLE II. End item classification of defects (Continued)

	Major	Minor
Components not free from defects which do not impair their serviceability.		X
Removable components cannot be removed and replaced without difficulty.	X	
Components not properly assembled or aligned.	X	
Chair not stable on floor (not level, contains rock motion).	X	
Upholstery		
Upholstery not as specified.	X	
Seams not flush.	X	
Permanently stained or spotted in a way that is clearly noticeable and that cannot be removed with a cleaning agent.	X	
Stained or spotted (not permanently) in a way that is noticeable but that can be easily removed with a cleaning agent.		X
Color not uniform seriously affecting appearance.	X	
Color not uniform, but not seriously affecting appearance.		X
Upholstery cut, ripped, punctured.	X	
Contains open scratches or abrasions having a serious effect on serviceability or appearance.	X	

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TABLE II. End item classification of defects (Continued)

	Major	Minor
Upholstery		
Contain scratches or abrasions not having a serious effect on serviceability or appearance.		X
Contains creases, or wrinkles seriously affecting serviceability or appearances.	X	
Contains creases, or wrinkles not seriously affecting serviceability or appearances.		X
Upholstery not same on any one component or does not match any other component of the module.	X	
Upholstery not free of objectionable odor (aromatic orders usually associated with specific types of material shall not be regarded as objectionable).	X	
Upholstery patched or mended (seriously affecting serviceability or appearance).	X	
Heat Seals (where applicable)		
Seam blistered, cracked, or creased.		X
Seam not continuous or secure.	X	
Seam open.	X	
Stitching (where applicable)		
Loose or broken stitching resulting in poorly secured seam.	X	
Tight stitching resulting in rupture of upholstery.	X	
Open seams.	X	
Seams not straight or evenly spaced.		X

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TABLE II. End item classification of defects (Continued)

	Major	Minor
Stitching (where applicable)		
Seam pleated, puckered badly, or twisted.	X	
Needle chews (affecting appearance seriously).	X	
Needle chews (not affecting appearance seriously).		X
Raw edge of material noticeable at seam.		X
Stitching		
Thread ends not trimmed off neatly (seriously affecting appearance).	X	
Thread ends not trimmed off neatly (not seriously affecting appearance).		X
Damaged stitching or seam after installation (end item) which seriously affects appearance or serviceability.	X	
Damaged stitching or seam after installation (end item) which does not seriously affect appearance or serviceability.		X
Identification Marking		
Identification marking missing, incomplete, not legible.		X
Identification marking incorrect or not permanent.		X
Caution plates (where applicable) missing, incomplete, not legible.		X
Caution plates (where applicable) not permanently secured.		X
Marking not properly located.		X

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TABLE II. End item classification of defects (Continued)

	Major	Minor
Service Data		
Service data not included.	X	

4.4.3 Dimensional examination. Dimensional examination shall be conducted for compliance with dimensions specified. Any dimension not within specification tolerance shall be a defect. The inspection level shall be S-3 with an AQL of 1.0. Unit of product shall be the component of an end item or an end item, as applicable.

4.4.4 In-process testing. In-process testing shall be conducted to ascertain components or assemblies which cannot be readily tested in the end item are in accordance with specification requirements.

4.4.5 End item testing. End item testing shall be conducted in accordance with MIL-STD-105 and as indicated in table III, except that the acceptance number shall be zero for all sample sizes. A failure during 100 percent inspection shall cause rejection of that component or item only (not the entire lot). Unit of product shall be one chair.

TABLE III. Sampling for test

Characteristic	Requirement	Test Procedure	Inspection Level
Chair length and seat height	3.3.3.2	4.4.5.2.1	S-2
Armrest test	3.3.3.3	4.4.5.2.10	S-2
Headrest test	3.3.3.4	4.4.5.2.9	S-2
Seat height adjustments	3.3.4.1	4.4.5.2.4	S-2
Chair rotation test	3.3.4.2	4.4.5.2.5	S-2
Trendelenberg position test	3.3.4.3	4.4.5.2.7	S-2

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TABLE II. End item classification of defects (Continued)

Characteristic	Requirement	Test Procedure	Inspection Level
Chair positioning test	3.3.4.4	4.4.5.2.3	S-2
Chair operation time	3.3.4.6	4.4.5.2.6	S-2
Toe section loading test	3.3.5.2	4.4.5.2.2	S-2
Chair operational test	3.3.9	4.4.5.2.8	100%

4.4.5.1 Raw materials. Sufficient amount from each lot of material used in the construction of each lot shall be selected for chemical analysis.

4.4.5.2 Tests.

4.4.5.2.1 Chair length and seat width. The seat width shall be measured between the armrests at the narrowest point. The length of the chair shall be measured from the top of the headrest to end of the toeboard with the chair in the supine position with the headrest fully retracted.

4.4.5.2.2 Toe section loading test. A minimum weight of 220 pounds shall be placed upon the center of the free end of the toe section with the toe section horizontal (parallel to floor) with the chair in its highest position for a minimum of 5 minutes. The chair base assembly shall be observed for tipping. After removal of the weight, the chair shall be examined for structural damage.

4.4.5.2.3 Chair positioning test. Shall be measured with the headrest removed and the backrest in the fully raised position by placing a rigid bar along the backrest and measuring the angle between that line and a line parallel to the floor.

4.4.5.2.4 Seat height adjustment. The adjustable seat height measurement shall be made with the upholstery in place. Measurement shall be from the bottom of the floor plate to a line horizontally extended from the top of the seat cushion where a line extended from the backrest intersects the seat.

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4.4.5.2.5 Chair rotation test. The chair shall be rotated in the horizontal plane and the angular range of rotation in the both left and right shall be measured. The chair shall be stopped at random position and the extreme position and locked at each position.

4.4.5.2.6 Chair operation time. The time to operate the chair from the fully supine to the fully upright position and from the fully lowered to the fully raised position shall be measured. The measurement shall be repeated for the reverse sequences. A stop watch or an equivalent timing device with an accuracy of plus or minus one second or better shall be used.

4.4.5.2.7 Trendelenberg position test. The chair shall been in the modified trandelenberg position. Measurement shall be made with the headrest removed and the backrest in the fully lowered position by placing a rigid bar along the backrest and measuring the angle between that line and a line pa rallel to the floor.

4.4.5.2.8 Chair operation test. The chair shall be subjected to operational testing which shall consist of, but shall not be limited to, the following systems and components. Each system shall be operated for a minimum of three operations or cycles (as applicable).

a. Raising and lowering of chair including operation of speed control.
b. Chair controls (positioning control switches and foot controls).
c. Operation of arm assemblies including lock mechanism.
d. Head rest (attachment, removal and adjustability).
e. Emergency stop controls.
f. Left to right conversion capability without tools.
g. Pre-set automatic positioning and return.
h. Rotational and transverse capabilities including stop and locking capability.
i. Operation from fully upright to supine position (and trendelenberg position).

4.4.5.2.9 Headrest test. The loaded headrest shall be tested in accordance with the headrest test as specified in ANSI/ADA Specification No. 46.

4.4.5.2.10 Armrest test. The armrest shall be tested in accordance with the armrest load test as specified in ANSI/ADA Specification No. 46.

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5. PACKAGING

5.1 Preservation. Preservation shall be level C.

5.1.1 Level C. Each dental chair shall be packaged and cushioned in standard commercial containers of a suitable size and design that will afford adequate protection against deterioration and damage during shipment and storgage.

5.2 Packing. Packing shall be level C.

5.2.1 Level C. Each packaged dental chair shall be packed in a substantial commercial exterior container (export when applicable) which will afford adequate protection to the contents during shipment from the supply source to the receiving activity. This pack shall conform to the carrier's rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Interior packages shall be marked to identify contents and quantity.

5.3.2 Exterior container. Each exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Options required (see 3.3.3.4, 3.3.7.3, 3.3.8.)
(c) First Article when required (see 3.2.1).
(d) Approval of colors (see 3.6).
(e) For Naval requisitions (see 3.3.7.1).
(f) Dental operatory light, when required (see 3.3).

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6.1.2 The following options are listed to allow the user to choose those items required to effectively treat patients at their facility. In each category several options are given. If no option is checked, the first listed shall be the default value.

a. Electrical requirements:

(1) 115 ± 10% volts, 60 ± 2 hertz
(2) 115 ± 10% volts, 50 ± 2 hertz
(3) 230 ± 10% volts, 50 ± 2 hertz
(4) 230 ± 10% volts, 60 ± 2 hertz

b. Headrest:

(1) Adjustable height, positive locking, articulating headrest
(2) Adjustable height, positive locking, non-articulating headrest
(3) Adjustable height, positive locking, prosthodontic (prosthetic) headrest

c. Upholstery color:

(1) Tan/beige
(2) Brown
(3) Rust
(4) Dark blue
(5) Light blue
(6) Gray
(7) Green
(8) Other

d. Seat cushion:

(1) Non-reversible
(2) Reversible

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1958

Location: NCR Sys Attdental

Civil Agency Coordinating Activities:
VA-OSS
PHS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-42005A(DM)	CHAIR, DENTAL PATIENT

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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